

Contents

1. Introduction to Combinatorics and Syllabus	3
1.1 What is Combinatorics?	3
1.2 Applications in Computer Science	3
1.3 Syllabus of Unit II - Combinatorics	3
2. Basic Counting Principles	4
2.1 The Product Rule	4
2.2 The Sum Rule	4
2.3 The Subtraction Rule (Inclusion-Exclusion)	4
2.4 The Division Rule	4
2.5 The Pigeonhole Principle	5
3. Permutations and Combinations	6
3.1 Permutations	6
3.2 Combinations	6
3.3 Enumerating Arrangements (Circular, Repetition, Multiset)	7
3.4 Selections with Repetition (Stars and Bars)	7
3.5 Binomial Coefficients and the Binomial Theorem	7
4. Principle of Inclusion-Exclusion	8
4.1 Two and Three Sets	8
4.2 General Form	8
5. Derangements	9
6. Generating Functions	10
6.1 Definition	10
6.2 Standard Series and Examples	10
7. Recurrence Relations	11
7.1 Definition, Order and Degree	11
7.2 Linear, Homogeneous and Non-homogeneous	11
8. Solving Linear Recurrence Relations	12
8.1 The Characteristic Equation	12
8.2 Distinct, Repeated and Complex Roots	12
8.3 Particular Solutions	13
9. Solving Recurrences Using Generating Functions	14

10. Question Bank - Unit II (Solved)	15
11. Quick Revision Summary	18

1. Introduction to Combinatorics and Syllabus

1.1 What is Combinatorics?

Combinatorics is the branch of mathematics that deals with **counting**, arranging and selecting objects. Every question in this unit has the form “in how many ways...?”. Whereas calculus studies continuous change, combinatorics studies **discrete** (separate, countable) structures such as finite sets, arrangements and sequences.

Combinatorics gives us tools to count without listing: the product and sum rules, permutations, combinations, the principle of inclusion-exclusion, generating functions and recurrence relations.

1.2 Applications in Computer Science

- Analysis of algorithms (counting the number of steps or operations).
- Probability and statistics (counting favourable outcomes).
- Coding theory and error-correcting codes.
- Cryptography (counting keys and passwords).
- Database and network design (counting paths, configurations).

1.3 Syllabus of Unit II - Combinatorics

Unit II (08 Hrs):

Combinations and Permutations, Enumerating combinations and permutations, Principle of Inclusion-Exclusion and Derangements. Generating functions, Recurrence Relations, Solution of Recurrence Relations.

The unit is organised as: basic counting principles, permutations and combinations, inclusion-exclusion, derangements, generating functions, recurrence relations and their solutions.

2. Basic Counting Principles

2.1 The Product Rule

Product Rule: If a procedure can be broken into a sequence of two tasks, with n_1 ways to do the first and n_2 ways to do the second, then there are $n_1 \cdot n_2$ ways to do the procedure. For k tasks the count is $n_1 n_2 \dots n_k$.

Example 1: Two routes join city A to city B and three routes join B to C. A journey from A to C can be made in $2 \times 3 = 6$ ways.

Example 2: Licence plates with three uppercase letters followed by three digits: $26^3 \times 10^3 = 17,576,000$.

2.2 The Sum Rule

Sum Rule: If a task can be done in one of n_1 ways *or* in one of n_2 ways, and no way is common to both, then there are $n_1 + n_2$ ways. In set form, $|A \cup B| = |A| + |B|$ when A and B are disjoint.

Example 1: A department chooses a student or a faculty member as representative; there are 37 faculty and 83 majors and no overlap, so $37 + 83 = 120$ choices.

Example 2: Programming labels that are either a single letter or a letter followed by a digit: $26 + 26 \times 10 = 286$.

2.3 The Subtraction Rule (Inclusion-Exclusion)

Subtraction Rule: If a task can be done in n_1 ways or n_2 ways, the total is $n_1 + n_2$ *minus* the number of ways common to both:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

This is the simplest form of the principle of inclusion-exclusion.

2.4 The Division Rule

Division Rule: If a task can be done in n ways and every distinct outcome is counted the same number d of times, then the number of distinct outcomes is n/d .

Example: Seating 5 people in a circle is counted as $5!$ linear orders, but each circular arrangement is counted 5 times (one per rotation), so the number is $5!/5 = 4! = 24$.

2.5 The Pigeonhole Principle

Pigeonhole Principle: If $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Generalized form: If N objects are placed into k boxes, then at least one box contains at least $\lceil N/k \rceil$ objects.

Example 1: Among any 367 people at least two share a birthday (367 objects, 366 possible birthdays).

Example 2: Among 100 people, at least $\lceil 100/12 \rceil = 9$ were born in the same month.

Example 3: If 6 colours are used to paint 37 homes, some colour is used on at least $\lceil 37/6 \rceil = 7$ homes.

3. Permutations and Combinations

3.1 Permutations

Definition: A **permutation** of a set of distinct objects is an *ordered* arrangement of them. An ordered arrangement of r elements is called an **r -permutation**. The number of r -permutations of a set with n elements is denoted $P(n,r)$.

Formula: For $1 \leq r \leq n$:

$$P(n, r) = n(n-1)(n-2)\dots(n-r+1) = n! / (n-r)!$$

In particular, the number of permutations of all n objects is $P(n,n) = n!$

Example 1: For $S = \{1,2,3\}$ the 2-permutations are 12, 13, 21, 23, 31, 32, so $P(3,2) = 6$.

Example 2: First, second and third prize from 100 people: $P(100,3) = 100 \cdot 99 \cdot 98 = 970,200$.

Example 3: A saleswoman visits 8 cities starting at a fixed city: $7! = 5040$ orders.

Example 4 (block method): Permutations of ABCDEFGH containing the string ABC — treat ABC as one block, giving 6 objects: $6! = 720$.

3.2 Combinations

Definition: An **r -combination** of a set is an *unordered* selection of r elements, i.e. a subset of size r . Its number is $C(n,r)$ or $\binom{n}{r}$, called the **binomial coefficient**.

Formula: For $n \geq r \geq 0$:

$$C(n, r) = n! / (r! (n-r)!)$$

Also $C(n,r) = C(n, n-r)$, and $C(n,r) = P(n,r)/r!$ (each subset has $r!$ orderings).

Example 1: $C(4,2) = 6$; the 2-combinations of $\{a,b,c,d\}$ are $\{a,b\}$, $\{a,c\}$, $\{a,d\}$, $\{b,c\}$, $\{b,d\}$, $\{c,d\}$.

Example 2: Poker hands of 5 cards from 52: $C(52,5) = 2,598,960$; also $C(52,47) = C(52,5)$.

3.3 Enumerating Arrangements (Circular, Repetition, Multiset)

Useful formulas:

- Arrangements of n distinct objects in a row: $n!$
- Circular arrangements of n objects: $(n-1)!$
- r -permutations with repetition allowed (from n types): n^r
- Permutations of a multiset with $n_1, n_2, \dots$ repeated objects: $n! / (n_1! n_2! \dots)$

Example 1: Five people around a round table: $(5-1)! = 24$.

Example 2: Arrangements of the letters of LEVEL (L twice, E twice, V once): $5! / (2! \cdot 2!) = 30$.

3.4 Selections with Repetition (Stars and Bars)

Formula: The number of r -combinations with repetition from n types, equivalently the number of non-negative integer solutions of $x_1 + x_2 + \dots + x_n = r$, is

$$\binom{n+r-1}{r}$$

If each $x_i \geq c_i$, first substitute $x_i' = x_i - c_i \geq 0$.

Example 1: Solutions of $x + y + z = 17$ (all ≥ 0): $C(19,17) = C(19,2) = 171$.

Example 2: Solutions of $x + y + z = 17$ with $x \geq 1, y \geq 2, z \geq 3$: put $x' = x-1, y' = y-2, z' = z-3$ so $x' + y' + z' = 11$, giving $C(13,11) = 78$.

Example 3: Solutions of $x_1 + x_2 + x_3 = 10$ with $x_1 \geq 2$: $C(10,8) = 45$.

3.5 Binomial Coefficients and the Binomial Theorem

Identities:

Pascal's identity: $C(n+1, k) = C(n, k) + C(n, k-1)$.

Binomial theorem: $(x + y)^n = \sum_{k=0}^n C(n, k) x^{n-k} y^k$.

Special sums: $\sum C(n, k) = 2^n$ and $\sum (-1)^k C(n, k) = 0$.

Example: $C(4,2) = C(3,2) + C(3,1) = 3 + 3 = 6$; and $\sum_k C(3, k) = 1+3+3+1 = 8 = 2^3$.

4. Principle of Inclusion-Exclusion

4.1 Two and Three Sets

Formulas:

Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$.

Three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example: Integers from 1 to 500 divisible by 3 or 5 or 7. With $\lfloor 500/3 \rfloor = 166$, $\lfloor 500/5 \rfloor = 100$, $\lfloor 500/7 \rfloor = 71$, pairwise $\lfloor 500/15 \rfloor = 33$, $\lfloor 500/21 \rfloor = 23$, $\lfloor 500/35 \rfloor = 14$, triple $\lfloor 500/105 \rfloor = 4$:

$$|A \cup B \cup C| = 166 + 100 + 71 - 33 - 23 - 14 + 4 = 271$$

4.2 General Form

General Inclusion-Exclusion:

$$|A_1 \cup \dots \cup A_n| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|$$

Example (divisible by 3 and 5 but not 7): Divisible by 15: $\lfloor 500/15 \rfloor = 33$. Divisible by 3, 5 and 7 (i.e. by 105): $\lfloor 500/105 \rfloor = 4$. So the answer is $33 - 4 = 29$.

5. Derangements

Definition: A **derangement** is a permutation of objects in which **no** object occupies its original position. The number of derangements of n objects is denoted D_n (or $!n$).

Formula and recurrence:

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right) = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

Recurrence: $D_n = (n-1)(D_{n-1} + D_{n-2})$, equivalently $D_n = n D_{n-1} + (-1)^n$.

Values: $D_1=0$, $D_2=1$, $D_3=2$, $D_4=9$, $D_5=44$.

Example: Derangements of $\{1,2,3,4\}$: $D_4 = 24(1 - 1 + 1/2 - 1/6 + 1/24) = 24 \times (9/24) = 9$.

6. Generating Functions

6.1 Definition

Definition: The **generating function** for the sequence $a_0, a_1, a_2, \dots$ is the infinite series

$$G(x) = a_0 + a_1x + a_2x^2 + \dots + a_rx^r + \dots = \sum_{r=0}^{\infty} a_r x^r$$

For a finite sequence $a_0, \dots, a_n$: $G(x) = \sum_{r=0}^n a_r x^r$.

6.2 Standard Series and Examples

Standard generating functions:

Sequence	Generating function
1, 1, 1, 1, ...	$1 / (1 - x)$
1, a, a^2 , a^3 , ...	$1 / (1 - ax)$
1, -1, 1, -1, ...	$1 / (1 + x)$
1, 2, 3, 4, ... ($a_r = r+1$)	$1 / (1 - x)^2$
$a_r = C(r+2, 2)$	$1 / (1 - x)^3$
$C(n, 0), C(n, 1), \dots$	$(1 + x)^n$

Examples:

- $a_r = 3 \rightarrow 3 / (1 - x)$
- $a_r = r + 1 \rightarrow 1 / (1 - x)^2$
- $a_r = 2^r \rightarrow 1 / (1 - 2x)$
- Sequence 0, 1, 2, 3, ... ($a_r = r$) $\rightarrow x / (1 - x)^2$

7. Recurrence Relations

7.1 Definition, Order and Degree

Definition: A **recurrence relation** for a sequence a_n is an equation that expresses a_n in terms of one or more previous terms of the sequence. With initial conditions it determines the whole sequence. Example: $a_n = a_{n-1} + 3$, $n \geq 2$, with $a_1 = 5$ gives 5, 8, 11, 14, ...

Order and Degree:

- **Order** = difference between the largest and smallest subscripts. E.g. $a_n - 3a_{n-1} + 2a_{n-2} = 0$ has order 2.
- **Degree** = highest power of a_n . E.g. $a_n^2 + a_{n-1}^2 = 1$ has degree 2.

7.2 Linear, Homogeneous and Non-homogeneous

Definitions:

- **Linear:** a recurrence of degree one, of the form $c_0a_n + c_1a_{n-1} + \dots + c_ka_{n-k} = f(n)$ with constant coefficients.
- **Homogeneous:** $f(n) = 0$.
- **Non-homogeneous:** $f(n) \neq 0$.

8. Solving Linear Recurrence Relations

8.1 The Characteristic Equation

Method: The general solution is

$$a_n = a_n^h + a_n^p$$

where a_n^h (homogeneous part) is found by setting $f(n) = 0$ and solving the **characteristic equation**

$$c_0\alpha^k + c_1\alpha^{k-1} + \dots + c_k = 0$$

and a_n^p is a particular solution.

8.2 Distinct, Repeated and Complex Roots

Three cases for a_n^h :

- **Distinct roots** $\alpha_1, \dots, \alpha_k$: $a_n^h = A_1\alpha_1^n + \dots + A_k\alpha_k^n$
- **Repeated root** α of multiplicity m : $a_n^h = (A_1 + A_2n + \dots + A_m n^{m-1})\alpha^n$
- **Complex roots** $a \pm ib$: $a_n^h = A_1(a+ib)^n + A_2(a-ib)^n$

Example (distinct): Solve $a_n - 6a_{n-1} + 8a_{n-2} = 0$. Characteristic $\alpha^2 - 6\alpha + 8 = 0 \Rightarrow \alpha = 2, 4$, so $a_n^h = A_1 2^n + A_2 4^n$.

Example (with conditions): $a_n - 5a_{n-1} + 6a_{n-2} = 0$, $a_0=2$, $a_1=5$: $\alpha = 2, 3$; $a_n = 2^n + 3^n$.

Example (repeated): $a_n - 6a_{n-1} + 9a_{n-2} = 0$: $(\alpha-3)^2 = 0$, so $a_n^h = (A_1 + A_2n)3^n$.

Example (Fibonacci): $a_n = a_{n-1} + a_{n-2}$, $a_0=1$, $a_1=1$: $\alpha^2 - \alpha - 1 = 0 \Rightarrow \alpha = (1 \pm \sqrt{5})/2$, giving Binet's formula.

8.3 Particular Solutions

Cases for a_n^p (by the form of $f(n)$):

- **Case (i):** $f(n)$ a polynomial of degree $r \Rightarrow$ try $A + Bn + Cn^2 + \dots + En^r$.
- **Case (ii):** $f(n) = \beta^n$, β not a root $\Rightarrow$ try $A\beta^n$.
- **Case (iii):** $f(n) = (\text{poly}) \cdot \beta^n$, β not a root $\Rightarrow$ try $(\text{poly})\beta^n$.
- **Case (iv):** $f(n) = (\text{poly}) \cdot \beta^n$, β a root of multiplicity $(m-1) \Rightarrow$ try $n^{m-1}(\text{poly})\beta^n$.

Example (Case i): Solve $a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$. Characteristic $\alpha^2 + 5\alpha + 6 = 0 \Rightarrow \alpha = -2, -3$. Try $a_n^p = A + Bn + Cn^2$. Solving gives $C = 1/4$, $B = 17/24$, $A = 115/288$.

Example (Case ii): Solve $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$. Since 7 is not a root, try $A \cdot 7^n$; solving gives $A = 49/20$.

Example (Case iii): Solve $a_n - a_{n-1} = 3n \cdot 2^n$. Try $(A + Bn)2^n$; solving gives $A = -6$, $B = 6$.

Example (Case iv): Solve $a_n - 3a_{n-1} = 2 \cdot 3^n$. Since 3 is a root, try $An \cdot 3^n$; solving gives $A = 2$.

9. Solving Recurrences Using Generating Functions

Method:

1. Multiply the recurrence by x^r and sum over the valid range of r .
2. Express each sum in terms of $G(x)$, using the initial conditions.
3. Solve the resulting equation for $G(x)$.
4. Decompose $G(x)$ into partial fractions.
5. Read off a_r as the coefficient of x^r .

Example 1: Solve $a_r = 3a_{r-1}$, $a_0 = 2$. Summing gives $G(1 - 3x) = 2$, so $G = 2/(1-3x)$, hence $a_r = 2 \cdot 3^r$.

Example 2: Solve $a_r - 3a_{r-1} + 2a_{r-2} = 0$, $a_0=2$, $a_1=3$. Gives $G = 1/(1-x) + 1/(1-2x)$, so $a_r = 1 + 2^r$.

Example 3: Solve $a_r - 2a_{r-1} + a_{r-2} = 2^r$, $a_0=2$, $a_1=1$. Gives $G = 3/(1-x) - 5/(1-x)^2 + 4/(1-2x)$, so $a_r = 2^{r+2} - 5r - 2$.

Example 4: Solve $a_n = 3a_{n-1} + 2n$, $a_0=1$. Gives $a_n = (5/2)3^n - n - 3/2$.

10. Question Bank - Unit II (Solved)

Q1. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of (i) exactly 3 girls, (ii) at least 3 girls, (iii) at most 3 girls?

Solution. Total people = $9 + 4 = 13$.

- **(i) Exactly 3 girls:** $C(4,3) \cdot C(9,4) = 4 \times 126 = 504$.
- **(ii) At least 3 girls:** $C(4,3)C(9,4) + C(4,4)C(9,3) = 504 + 1 \times 84 = 588$.
- **(iii) At most 3 girls:** $C(13,7) - C(4,4)C(9,3) = 1716 - 84 = 1632$.

Q2. In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other?

Solution. Treat Alice and Bob as one block: 4 units $\rightarrow 4! = 24$; internal order $2! = 2$. Total = $24 \times 2 = 48$.

Q3. In how many ways can 7 people be seated such that two specific people are never adjacent?

Solution. Total = $7! = 5040$. Adjacent (block) = $6! \times 2! = 1440$. Never adjacent = $5040 - 1440 = 3600$.

Q4. How many integers between 1 and 500 (inclusive) are (a) divisible by at least one of 3, 5, 7? (b) divisible by 3 and 5 but not by 7?

Solution.

- **(a)** By inclusion-exclusion: $166 + 100 + 71 - 33 - 23 - 14 + 4 = 271$.
- **(b)** Divisible by 15: 33; divisible by 105: 4; so $33 - 4 = 29$.

Q5. How many integer solutions exist for $x_1 + x_2 + x_3 = 10$ where $x_1 \geq 2$ and $x_2, x_3 \geq 0$?

Solution. Put $x_1' = x_1 - 2 \geq 0$, so $x_1' + x_2 + x_3 = 8$. Number = $C(10,8) = C(10,2) = 45$.

Q6. Solve the recurrence relation $a_n = 3a_{n-1} + 2$, $a_0 = 1$, using any method.

Solution. Particular: $a_n^p = C \Rightarrow C = 3C + 2 \Rightarrow C = -1$. Homogeneous: $A \cdot 3^n$. So $a_n = A \cdot 3^n - 1$; $a_0 = A - 1 = 1 \Rightarrow A = 2$. Hence $a_n = 2 \cdot 3^n - 1$.

Q7. Find the solution of the recurrence relation $a_n - 3a_{n-1} = 2(3^n)$.

Solution. Homogeneous: $\alpha = 3$, so $a_n^h = A_1 3^n$. Since $\beta = 3$ is a root, take $a_n^p = A n \cdot 3^n$; substituting gives $A = 2$. Hence $a_n = (A_1 + 2n)3^n$.

Q8. Use a generating function to solve $a_n = 2a_{n-1} + 1$, $a_0 = 0$.

Solution. Summing gives $G(1 - 2x) = x/(1-x)$, so $G = x / [(1-x)(1-2x)] = -1/(1-x) + 1/(1-2x)$. Hence $a_n = 2^n - 1$.

Q9. Solve the recurrence $a_n = 3a_{n-1} - 2a_{n-2}$, $a_0 = 2$, $a_1 = 5$.

Solution. $\alpha^2 - 3\alpha + 2 = 0 \Rightarrow \alpha = 1, 2$, so $a_n = A + B \cdot 2^n$. $A + B = 2$, $A + 2B = 5 \Rightarrow B = 3$, $A = -1$. Hence $a_n = -1 + 3 \cdot 2^n$.

Q10. Use a generating function to solve $a_r - 3a_{r-1} + 2a_{r-2} = 0$, $a_0 = 2$, $a_1 = 3$.

Solution. Summing gives $G(1-x)(1-2x) = 2 - 3x$, so $G = 1/(1-x) + 1/(1-2x)$. Hence $a_r = 1 + 2^r$.

Q11. Solve $a_n - 2a_{n-1} - 3a_{n-2} = 0$, $a_0 = 3$, $a_1 = 1$, using a generating function.

Solution. Summing gives $G(1-2x-3x^2) = 3 - 5x$, so $G = 1/(1-3x) + 2/(1+x)$. Hence $a_n = 3^n + 2(-1)^n$.

Q12. Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$, for all $n \geq 2$, using a generating function.

Solution. Summing gives $G(1-3x)^2 = a_0 + (a_1 - 6a_0)x$, so $G = [a_0 + (a_1 - 6a_0)x]/(1-3x)^2$. Hence $a_n = (A + Bn)3^n$ (general solution).

Q13. Find the sequence b_r from the generating function $\sum b_r x^r = 1/(x^2 - 5x + 6)$.

Solution. $1/(x^2 - 5x + 6) = 1/[(x-2)(x-3)] = 1/(2-x) - 1/(3-x)$. Expanding, $b_r = 1/2^{r+1} - 1/3^{r+1} = (3^{r+1} - 2^{r+1})/6^{r+1}$.

Q14. (Short) Use the Pigeonhole Principle to prove: in any group of 13 people, at least two were born in the same month.

Solution. 12 months = 12 boxes, 13 people. Since $13 > 12$, some month contains at least two people.

Q15. (Short) How many permutations can be formed from the letters of the word LEVEL?

Solution. L twice, E twice, V once: $5!/(2! \cdot 2!) = 30$.

Q16. (Short) What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students?

Solution. Need $\lceil 30/k \rceil \geq 5$, i.e. $k < 7.5$, so the maximum is 7 groups.

Q17. (Short) Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$.

Solution. Particular $C = -1$, so $a_n = A \cdot 3^n - 1$; $a_0 = A - 1 = 2 \Rightarrow A = 3$. Hence $a_n = 3^{n+1} - 1$.

Q18. (Short) How many ways can 8 people be arranged in a line if two specific people must stand together?

Solution. $7! \times 2! = 5040 \times 2 = 10,080$.

Q19. (Short) How many ways can 5 people be seated in a circle?

Solution. $(5 - 1)! = 4! = 24$.

Q20. (Short) Solve the recurrence $a_n = 2a_{n-1}$, $a_0 = 1$.

Solution. $a_n = 2^n$.

Q21. (Short) Calculate the number of 3-combinations from a set of 5 distinct elements.

Solution. $C(5,3) = 10$.

11. Quick Revision Summary

Counting

- **Product rule:** stages multiply. **Sum rule:** disjoint options add. **Subtraction rule:** add and subtract the overlap. **Division rule:** divide out repeated counts.
- **Pigeonhole:** $k+1$ objects in k boxes $\rightarrow$ some box has ≥ 2 ; in general $\geq \lceil N/k \rceil$.

Permutations and Combinations

- $P(n,r) = n!/(n-r)!$; $C(n,r) = n!/(r!(n-r)!)$; $C(n,r) = P(n,r)/r!$.
- Circular = $(n-1)!$; repetition-permutations = n^r ; repetition-combinations = $C(n+r-1, r)$; multiset = $n!/(n_1!n_2!\dots)$.
- Binomial theorem: $(x+y)^n = \sum C(n,k) x^{n-k} y^k$; Pascal: $C(n+1,k) = C(n,k) + C(n,k-1)$.

Inclusion-Exclusion and Derangements

- $|A \cup B| = |A| + |B| - |A \cap B|$; extend to three sets with alternating signs.
- Derangements: $D_n = n! \sum (-1)^k / k!$; $D_n = (n-1)(D_{n-1} + D_{n-2})$; $D_4 = 9$.

Generating Functions

- $G(x) = \sum a_r x^r$. Key series: $1/(1-x)$, $1/(1-ax)$, $1/(1+x)$, $1/(1-x)^2 = \sum (r+1)x^r$, $1/(1-x)^3$.

Recurrence Relations

- Order, degree, linear, homogeneous/non-homogeneous.
- Solution $a_n = a_n^h + a_n^p$. Characteristic equation $c_0 \alpha^k + \dots + c_k = 0$: distinct $\rightarrow \sum A_i \alpha_i^n$; repeated $\rightarrow (A_1 + A_2 n + \dots) \alpha^n$; complex $\rightarrow a \pm ib$.
- Particular solutions: polynomial, β^n , $(\text{poly})\beta^n$, or $n^{m-1}(\text{poly})\beta^n$ when β is a root.
- Or solve via generating functions: multiply by x^r , sum, solve for $G(x)$, partial fractions, read coefficients.

End of Unit II - Combinatorics. Complete notes compiled from the lecture slides, the Unit-2 question bank and the worked recurrence-relation notes of Discrete Mathematics and Graph Theory (DMGT).